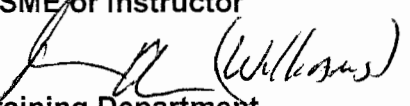


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM						
SYSTEM:	Residual Heat Removal						
TASK:	TCAF a Loss of RHR						
TASK NUMBER:	N1140300401						
JPM NUMBER:	11-01 NRC Sim d						
ALTERNATE PATH:	☐	K/A NUMBER:	APE 025 AA1.09				
APPLICABILITY:		IMPORTANCE FACTOR:	<table style="width: 100%; border: none;"><tr><td style="width: 50%; text-align: center;">3.2</td><td style="width: 50%; text-align: center;">3.1</td></tr><tr><td style="text-align: center;">RO</td><td style="text-align: center;">SRO</td></tr></table>	3.2	3.1	RO	SRO
3.2	3.1						
RO	SRO						
EO ☐	RO ☒	STA ☐	SRO ☒				
EVALUATION SETTING/METHOD:	Simulator						
REFERENCES:	S2.OP-AB.RHR-0001, Rev. 17 (Rev checked 8-6-12)						
TOOLS AND EQUIPMENT:	None						
VALIDATED JPM COMPLETION TIME:	<u>25 min</u>						
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>						
Developed By:	G Gauding Instructor	Date:	08-18-12				
Validated By:	Brennan SME or Instructor	Date:	08-24-12				
Approved By:	 Training Department	Date:	9/12/12				
Approved By:	 Operations Department	Date:	9/4/12				
ACTUAL JPM COMPLETION TIME:							
ACTUAL TIME CRITICAL COMPLETION TIME:							
PERFORMED BY:							
GRADE: ☐ SAT ☐ UNSAT							
REASON, IF UNSATISFACTORY:							
EVALUATOR'S SIGNATURE:			DATE:				

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Residual Heat Removal

TASK: TCAF a Loss of RHR

**TASK
NUMBER:** N1140300401

SIMULATOR SETUP: IC-254

MALF VL0626 22RH18 fails to position (0-100%) Final Value 100% (This is to mimic the 22RH18-A/S being C/T in the SHUT position).

RT-1 MALF RH0026A 21 RHR pump trip

RT-3 REM RH02A 22RH12 Initial -0 Final 100 Ramp 60 sec

RT-3 REM RH06A 22RH17 Initial -0 Final 100 Ramp 120 sec

INITIAL CONDITIONS:

Unit 2 is in MODE 4.

RCS pressure is 315 psig and stable.

RHR HX inlet temp is 289 degrees with a 10°F/hr cooldown in progress.

21 RHR loop is in service in shutdown cooling mode.

22 RHR loop is aligned for ECCS.

INITIATING CUE:

You are the Reactor Operator. Monitor the cooldown and report when 21 RHR HX inlet temp reaches 285°F.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Place 22 RHR loop in service in Shutdown Cooling (SDC) mode by:
 - A. Aligning RHR valves to SDC lineup. (Step 3.52 & Att. 2)
 - B. Starting 22 RHR pump. (Step 3.52 & Att. 2)
 - C. Establishing stable RHR flow of 1800-3000 gpm (Step 3.68)
 - D. Establishing RCS temperature stable or lowering (Step 3.68)

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
		Simulator Operator insert RT-1 on Evaluators direction. MALF: RH0026A 21 RHR Pump Trip			
			Responds to trip of 21 RHR pump by depressing the STOP PB.		
			Note to Evaluator: The PZR Hi lvl alarm on 2CC2 will annunciate shortly after the RHR pump is tripped. IF the candidate begins actions to respond to this alarm, THEN CUE that a second operator will refer to the Alarm Response Procedure for the PZR Hi level alarm.		
			Recognizes there is no RHR cooling, and enters S2.OP-AB.RHR-0001, Loss of RHR.		
	3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
	3.2	IF the RCS is vented to the Containment...	Recognizes the RCS is not vented to the containment and N/A's step.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
	3.3	Is RCS aligned for operation <101 ft. elevation (Reduced Inventory)?	Determines RCS is not aligned for operation <101 ft. elevation (Reduced Inventory) and goes to step 3.5.		
	3.5	Is the loss of RHR due to a mechanical failure or loss of electrical power to the in-service RHR pump?	Determines the loss of RHR is due to a mechanical failure or loss of electrical power to the in-service RHR pump and goes to step 3.50. Note: If operator answers NO here, then they will perform steps which are designed for system, rather than pump, problems, and is incorrect.		
	3.6	GO TO Step 3.50	Goes to Step 3.50		
	3.50	Is a heat sink available for Residual Heat Removal? • Component Cooling to RHR System • Service Water to Component Cooling System	Determines a heat sink available for Residual Heat Removal since SW and CCW are unaffected.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
	3.51	Is an RHR Loop available?	Determines a RHR Loop is available since 22 RHR loop is available (lined up in ECCS Mode).		
	3.52	PLACE the alternate RHR Loop in service: <u>IF</u> alternate loop is aligned for ECCS, THEN PERFORM Attachment 2, Aligning RHR Loop From ECCS To Shutdown Cooling. <u>OR</u> <u>IF</u> alternate RHR Loop is aligned for Shutdown Cooling, THEN PERFORM Attachment 3, Aligning RHR Loop For Shutdown Cooling.	Performance steps on next page		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
*	Att. 2 2.0	<u>IF</u> placing RHR Loop 22 in service, <u>THEN</u>: A. ENSURE RHR Loop 22 is aligned for ECCS injection. B. PLACE 22RH29 RHR PUMP MINIMUM FLOW VALVE in AUTO. C. Locally: • OPEN 22RH12 RHR HX BYPASS ISOLATION VALVE • OPEN 22RH17 RHR LETDOWN ISOLATION VALVE • OPEN 22RH18-A/S RHR LETDOWN ISOLATION VALVE AIR SUPPLY (Note: Time compression will be utilized for operators performing actions in the field.)	Determines 22 RHR loop is aligned for ECCS injection from initial conditions. Determines 22RH29, RHR PUMP MINIMUM FLOW VALVE is in AUTO. Directs a local operator to open 22RH12 RHR HX BYPASS ISOLATION VALVE, open 22RH17 RHR LETDOWN ISOLATION VALVE, and open 22RH18-A/S RHR LETDOWN ISOLATION VALVE AIR SUPPLY <u>CUE:</u> Time compression will be used to perform field actions. <u>Simulator Operator:</u> When directed to open 22RH18-A/S, delete MALF VL0626 22RH18 fails to position (0-100) (This allows valve to move when control console PBs are depressed.) <u>Then</u> insert RT-3 to open the 22RH12 and 22RH17. Report to operator when they are open. Also report that you have released the Red Blocking Tag and opened 22RH18-A/S, RHR LETDOWN ISOLATION VALVE AIR SUPPLY		This step to open 3 local valves is considered a critical task because failure to have these local valves opened will prevent placing 22 RHR loop in service in SDC.(No bypass or 22RH18 flow <u>control</u>)

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
			<u>Simulator Operator Cue:</u> IF requested for WCC to release tag, state that the NEO has the release for 22RH18- A/S LOCAL A/S TO 22RH18.		
		D. In the Control Room: • CLOSE 22RH18, RHR PUMP FLOW CONTROL VALVE • CLOSE 21RH18, RHR PUMP FLOW CONTROL VALVE • CLOSE 2RH20, RHR HX BYPASS • CLOSE 21CC16, RHR HX COMPONENT COOL OUT VALVE • OPEN 22CC16, RHR HX COMPONENT COOL OUT VALVE	Depresses CLOSE PB's for: • 22RH18 RHR PUMP FLOW CONTROL VALVE • 21RH18 RHR PUMP FLOW CONTROL VALVE • 2RH20 RHR HX BYPASS • 21CC16 RHR HX COMPONENT COOL OUT VALVE And verifies valves stroke closed. Depresses OPEN PB for 22CC16 and verifies it strokes open.		
		E. START 22 RHR PUMP	Depresses the START PB for 22 RHR pump and verifies the pump started.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
*		F. OPERATE 22RH18 and 2RH20, to maintain stable RHR flow to the Reactor Coolant System.	Adjusts 22RH18 RHR PUMP FLOW CONTROL VALVE and 2RH20 RHR HX BYPASS to establish stable RHR pump flow.		
		RETURN to procedure step in effect.	Returns to step 3.52 and initials step as complete.		
	3.53	Is RHR in service?	Determines RHR is in service by previous step performance.		
	3.54	Is a heat sink available for Residual Heat Removal? • Component Cooling to RHR System • Service Water to Component Cooling System	Determines a heat sink available for Residual Heat Removal since SW and CCW are unaffected.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
	3.55	Is RHR flow stable?	Checks RHR flow and determines flow is stable, and goes to Step 3.68 Note: Operator may adjust 22RH18 RHR PUMP FLOW CONTROL VALVE and 2RH20 RHR HX BYPASS if flow is not stable, or not 1800-3000 gpm per step 3.56. If this is the case, RHR system verification is performed at step 3.59. (It is the same as step 3.68).		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM Residual Heat Removal
TASK: TCAF Loss of RHR

	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	S / U	COMMENTS (Required for UNSAT Evaluation)
	3.68	Is RHR System normal as indicated by <u>ALL</u> of the following? • RHR Pumps – at least one running • RCS level- >97.5 ft. elevation <u>AND</u> stable <u>OR</u> rising consistent with current RCS makeup <u>AND</u> no excessive indication of RCS Leakage. • RHR flow – stable between 1800 and 3000 gpm • RCS temperature – stable or lowering	Verifies 22 RHR pump I/S Determines RCS is full (PZR level) Adds 21SJ49 RHR DISCH TO COLD LEGS flow and 22SJ49 RHR DISCH TO COLD LEGS flow to get 1800-3000 gpm. Uses P-250 computer to verify RCS temperature is stable or lowering. <u>IF</u> temperature is NOT stable or lowering, raises flow through HX by opening 22RH18 RHR PUMP FLOW CONTROL VALVE while lowering bypass flow by closing 2RH20 RHR HX BYPASS to turn RCS temperature. Note: It is acceptable to use either RCS temperature indication(s) OR RHR HX inlet temperature to determine if RCS temperature is stable or lowering. CETs will respond the quickest. After determining RCS temp is stable or lowering and goes to Step 4.0, state: “JPM is complete.”		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- JB* 1. Task description and number, JPM description and number are identified.
- JB* 2. Knowledge and Abilities (K/A) references are included.
- JB* 3. Performance location specified. (in-plant, control room, or simulator)
- JB* 4. Initial setup conditions are identified.
- JB* 5. Initiating and terminating Cues are properly identified.
- JB* 6. Task standards identified and verified by SME review.
- JB* 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- JB* 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 17 Date 8.24.12
- JB* 9. Pilot test the JPM:
 a. verify Cues both verbal and visual are free of conflict, and
 b. ensure performance time is accurate.
- N/A* 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: *JB Brennan*

Date: 8.24.12

SME/Instructor:

Date:

SME/Instructor:

Date:

OPERATOR TRAINING PROGRAM JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

Unit 2 is in MODE 4.
RCS pressure is 315 psig and stable.
RHR HX inlet temp is 289 degrees with a 10°F/hr cooldown in progress.
21 RHR loop is in service in shutdown cooling mode.
22 RHR loop is aligned for ECCS.

INITIATING CUE:

You are the Reactor Operator. Monitor the cooldown and report when 21 RHR HX inlet temp reaches 285°F.